ATENEO DE ZAMBOANGA UNIVERSITY

Transmission Media & Multiplexing

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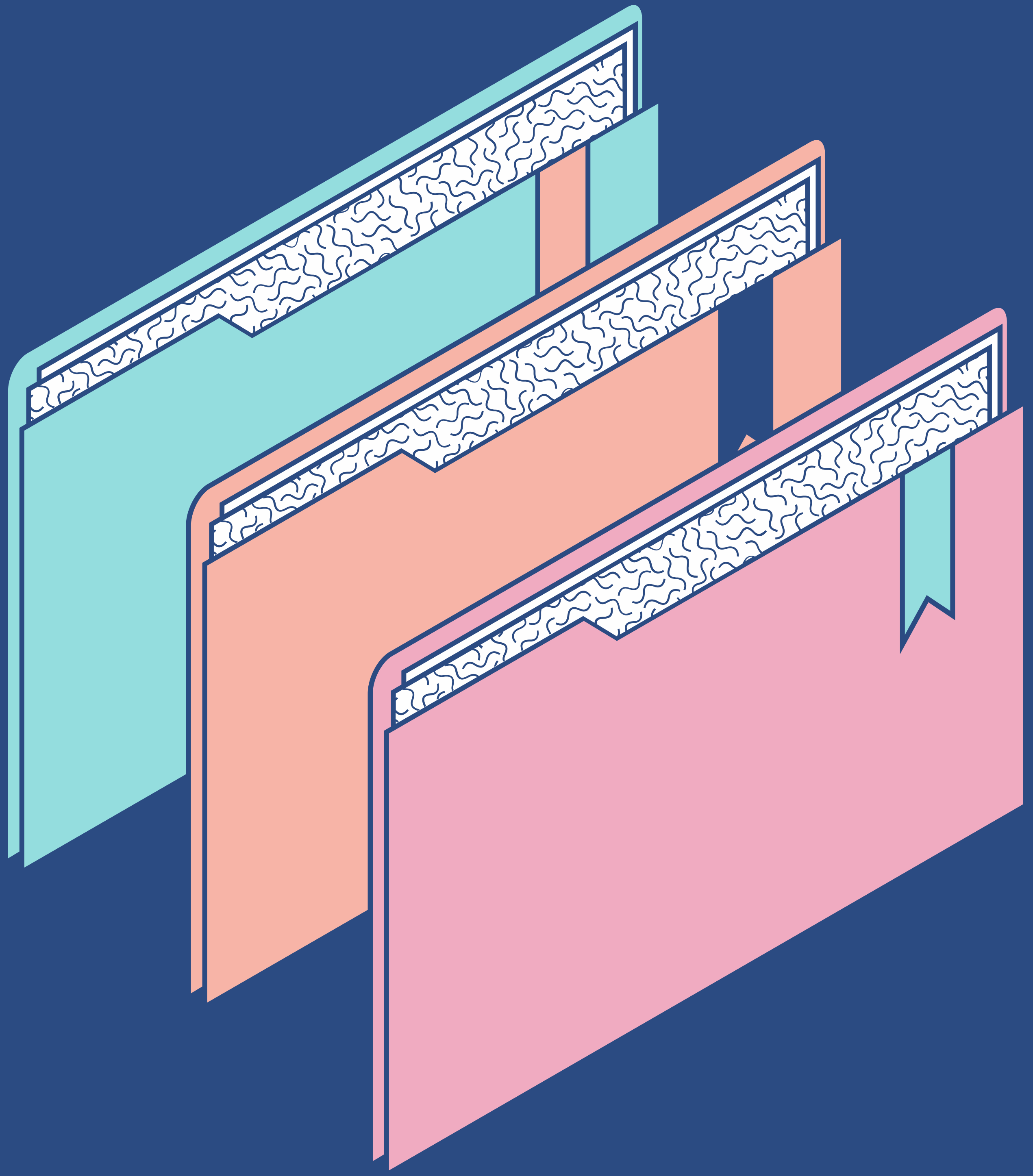
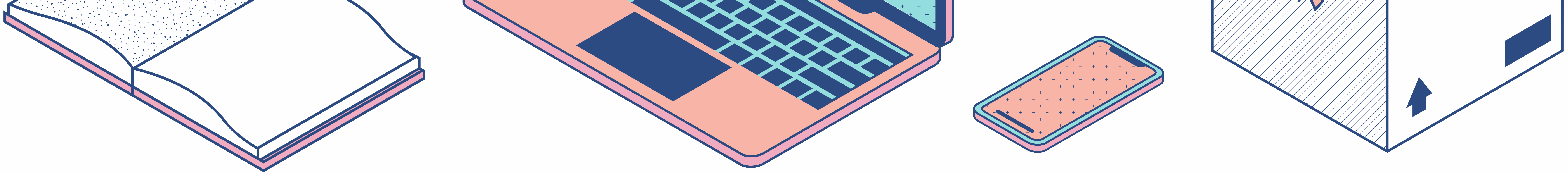
CIT.016 | DATA COMMUNICATION



Transmission Media



Transmission Medium

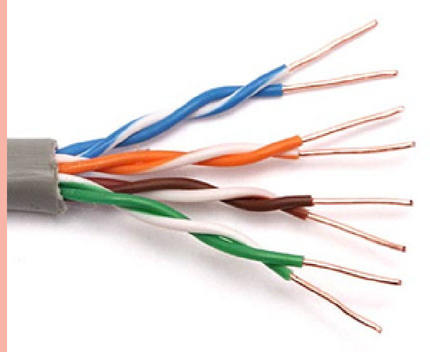
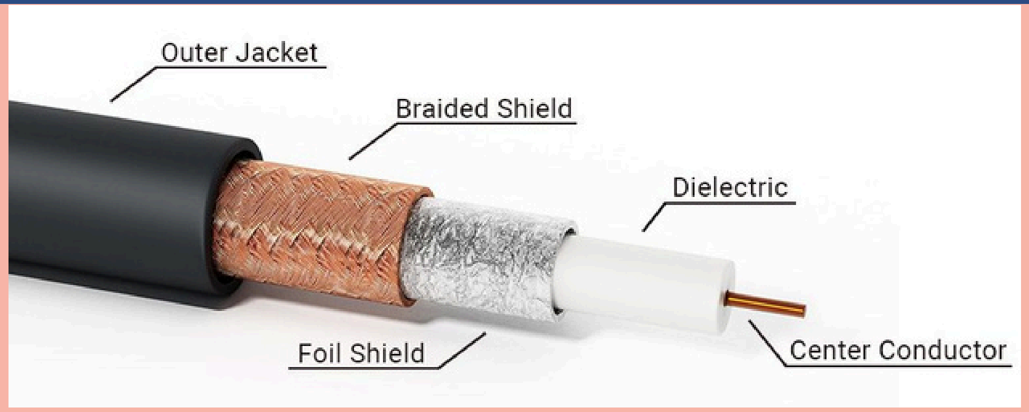
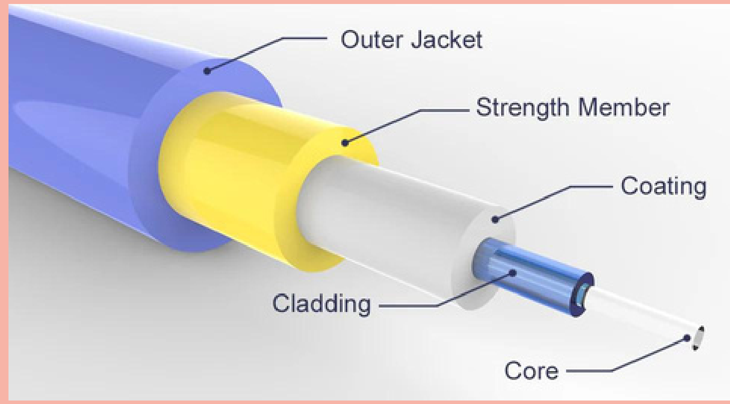
A transmission medium is a physical path between the transmitter and the receiver i.e. it is the channel through which data is sent from one place to another.

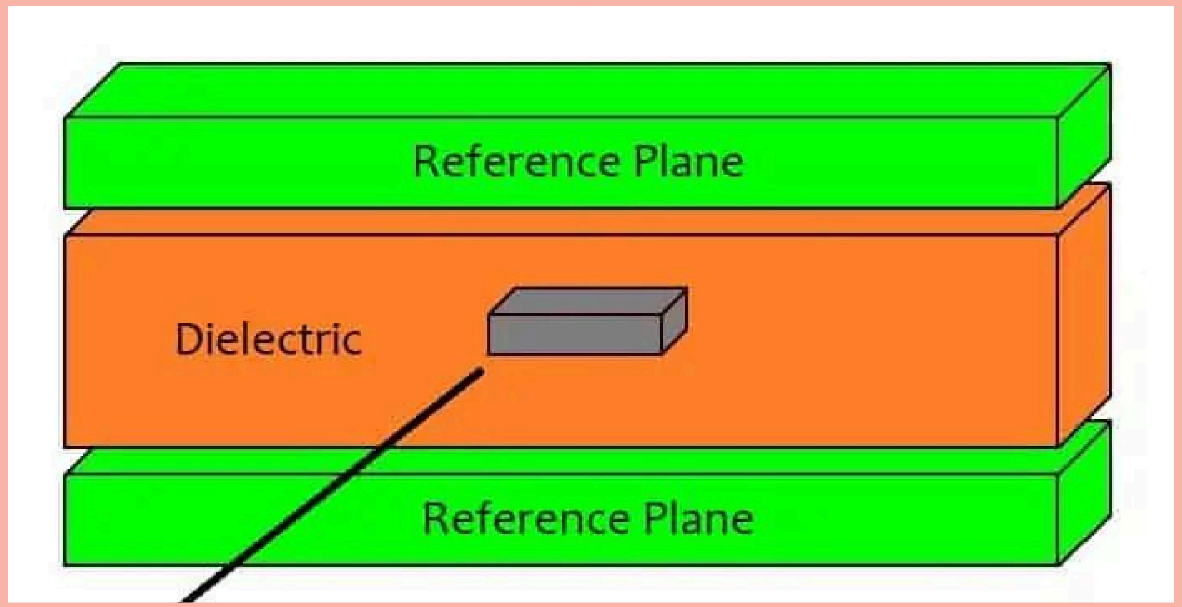
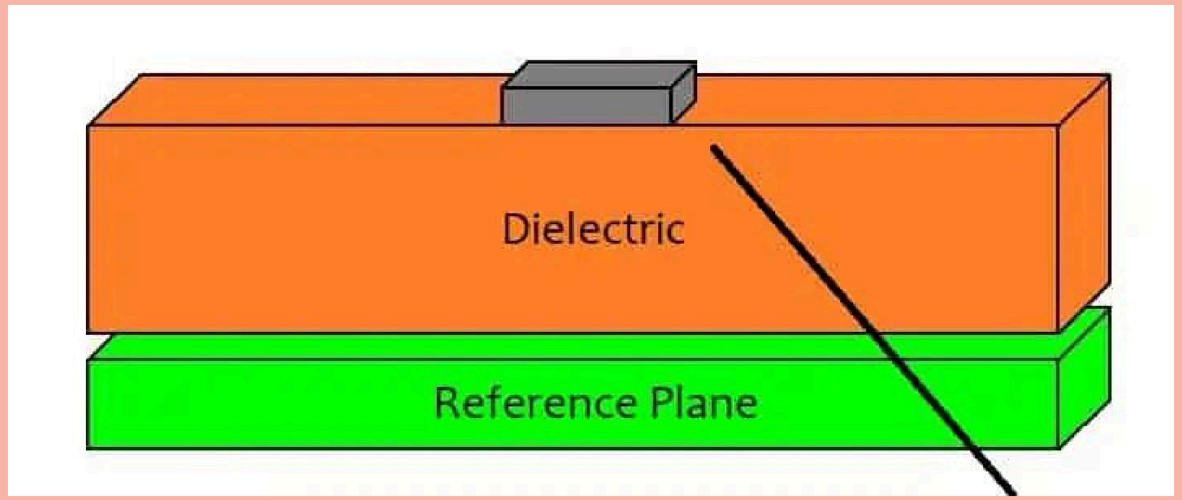
Guided Media

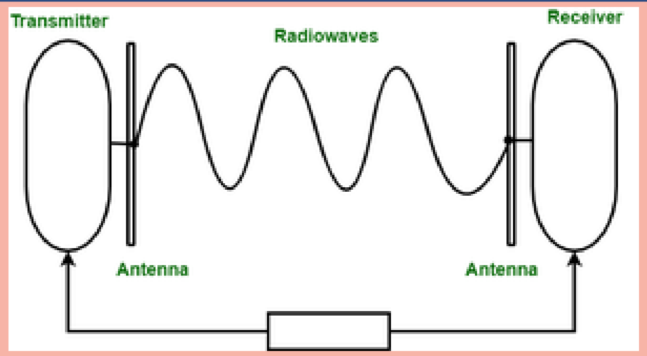
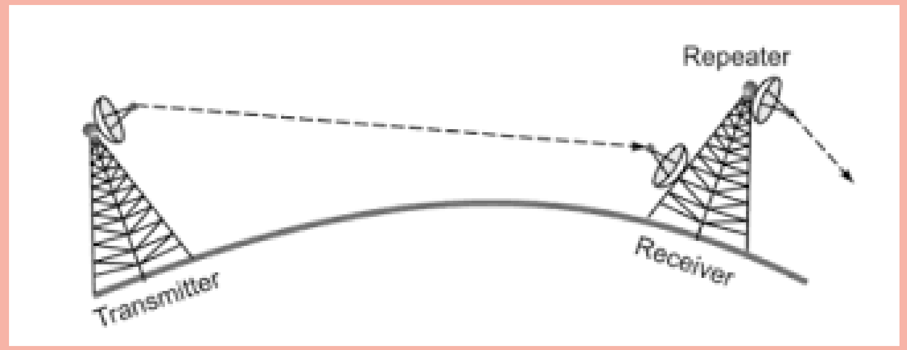
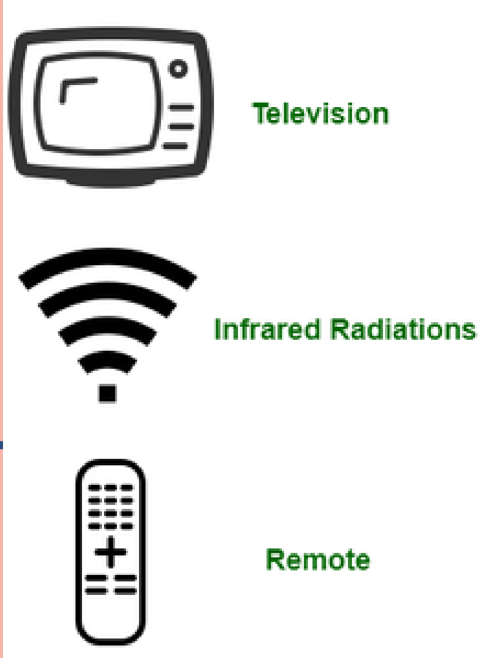
- **Wired** or **Bounded** transmission media.
- Signals being transmitted are directed and confined in a narrow pathway by using **physical links**.

Unguided Media

- **Wireless** or **Unbounded** transmission media.
- No physical medium is required for the transmission of **electromagnetic signals**.

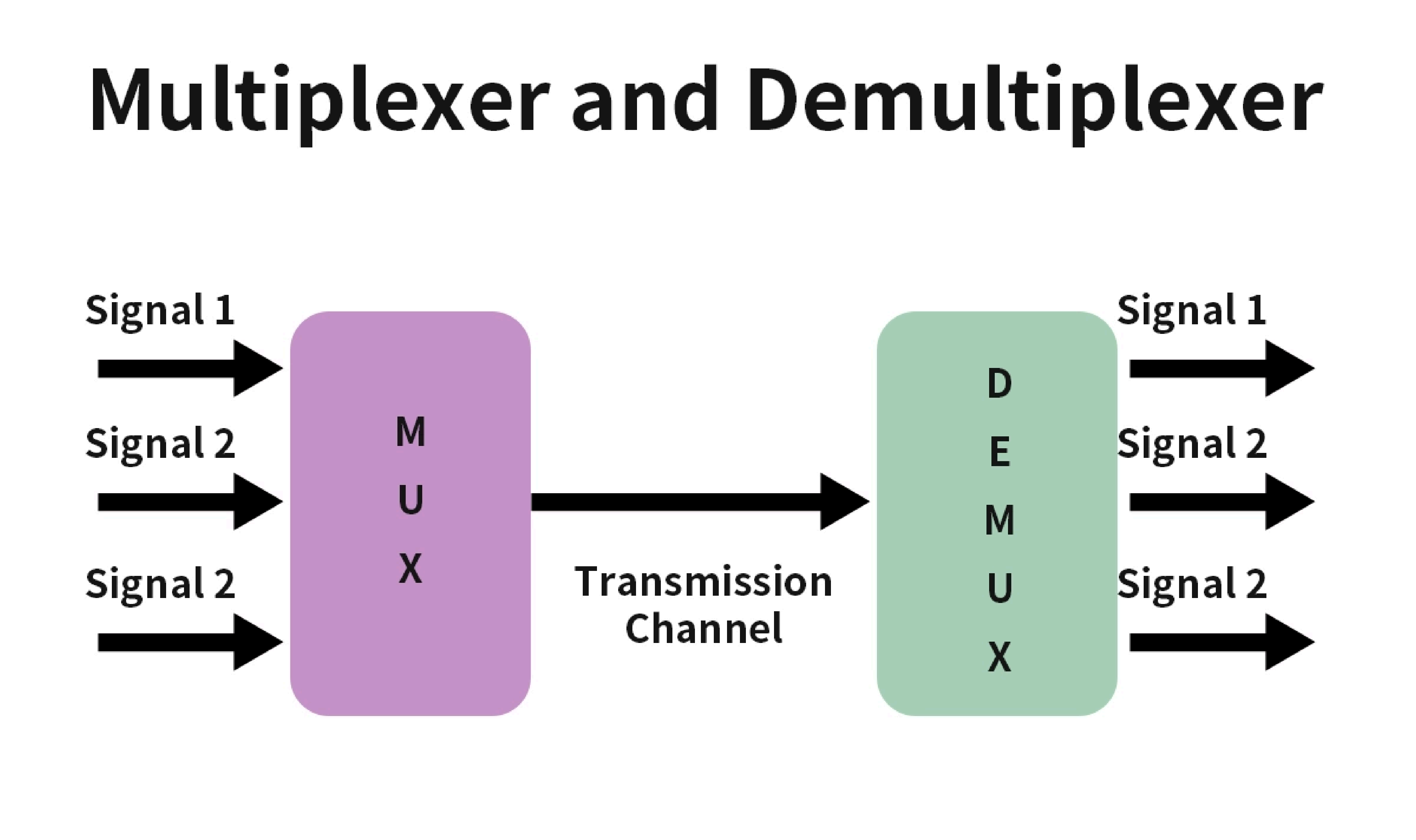
Guided Media	Definition	Illustration
Twisted Pair Cable	Consists of pairs of insulated copper wires twisted together. It is used in telephone networks and local area networks (LANs).	 
Coaxial Cable	Has a central conductor, an insulating layer, a metallic shield, and an outer insulating layer. It is used for cable television and broadband internet.	
Fiber-optic cable	Uses light to transmit data through glass or plastic fibers. It is used for high-speed data and long-distance communication.	

Guided Media	Definition	Illustration
Stripline	It uses a conducting material (waveguide) to transmit high-frequency waves, sandwiched between two layers of the ground plane which are usually shorted to provide EMI immunity.	
Microstripline	The conducting material is separated from the ground plane by a layer of dielectric.	

Unguided Media	Definition	Frequency Range	Illustration
Radio Waves	Used for wireless communication, such as Wi-Fi, Bluetooth, and radio broadcasting. It operates in various frequency ranges and can cover broad areas.	3KHz – 1GHz	
Microwaves	Utilized for point-to-point communication, such as satellite communication. It requires line-of-sight between transmitting and receiving antennas.	1GHz – 300GHz	
Infrared	Used for short-range communication, such as remote controls and certain wireless data transmission systems. It operates at a frequency just below visible light.	300GHz – 400THz	

Multiplexing

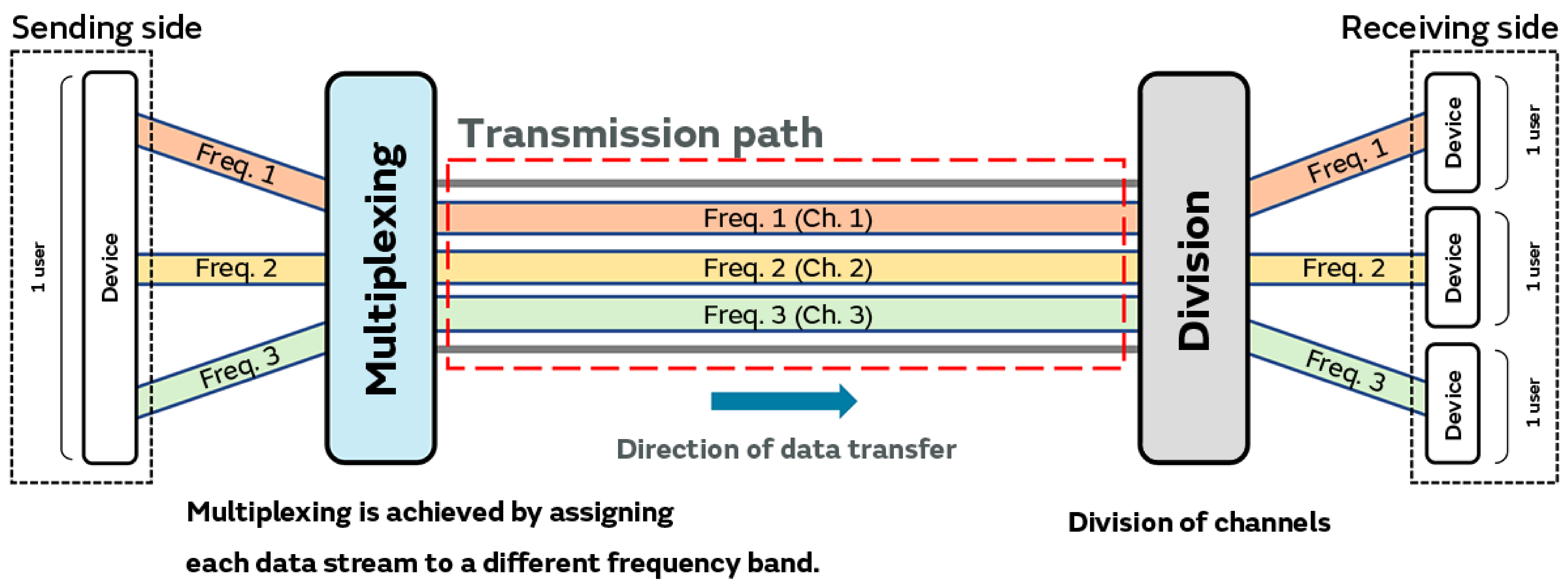
Multiplexing is the sharing of a medium or bandwidth. It is the process in which multiple signals coming from multiple sources are combined and transmitted over a single communication or physical line.



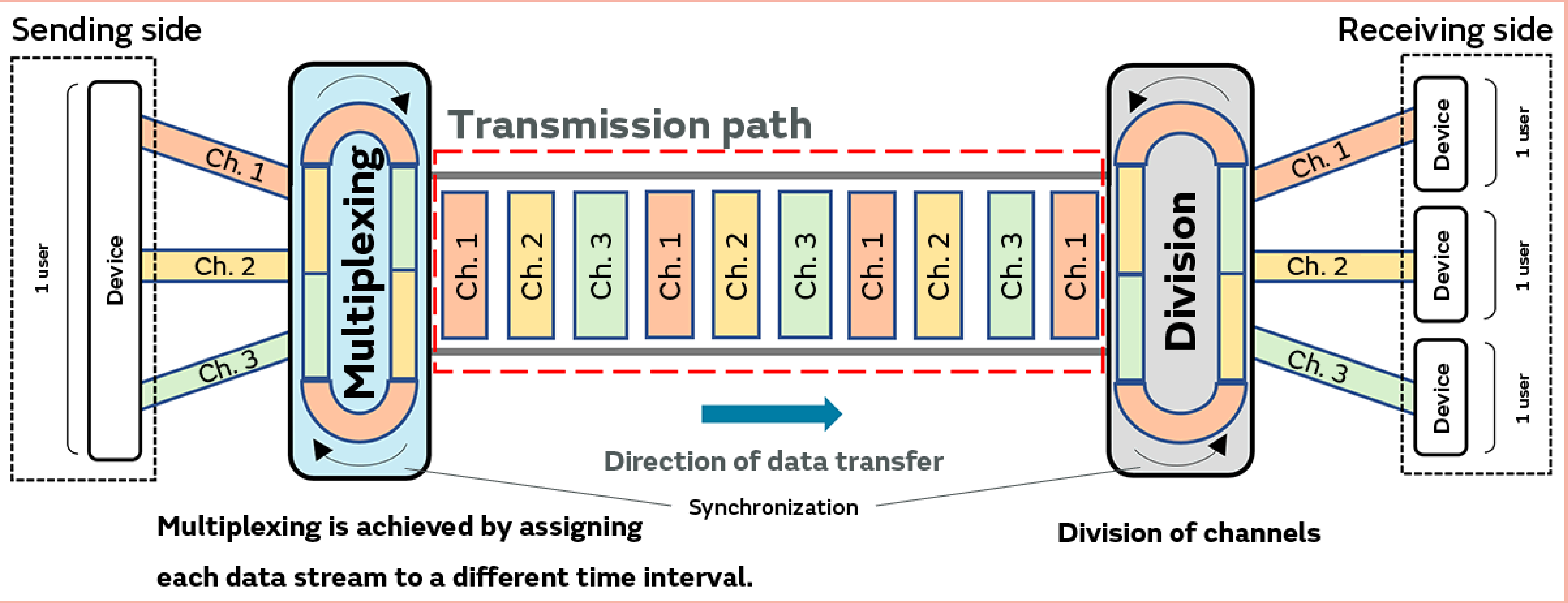
TYPES OF MULTIPLEXING

Multiplexing	Mode of Simultaneous Transmission of Multiple Signals
Frequency Division Multiplexing	over different frequency bands within the same communication channel
Time Division Multiplexing	share the same communication channel by taking turns and transmitting in different time slots
Wavelength Division Multiplexing	used in fiber optics; over different wavelengths (colors) of light within the same optical fiber.
Space-Division Multiplexing	over separate physical paths or channels, such as different optical fibers or wires
Code-Division Multiplexing	over the same frequency channel by using unique codes to distinguish each signal.

Frequency Division Multiplexing



Time Division Multiplexing



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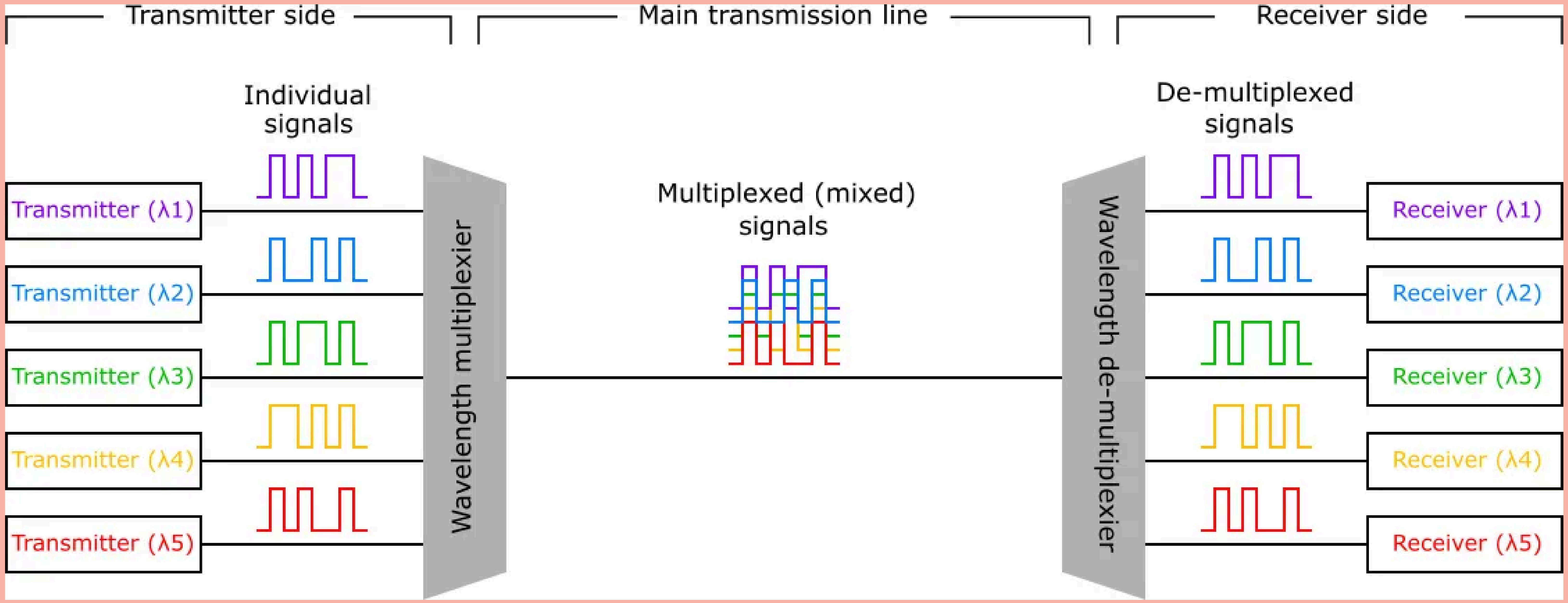
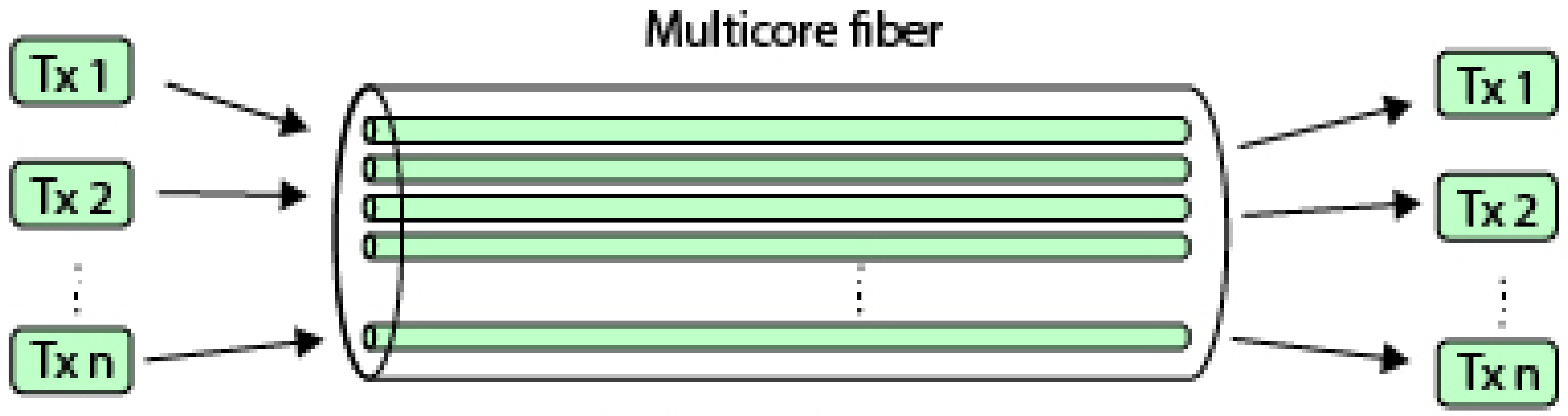
Wavelength Division Multiplexing

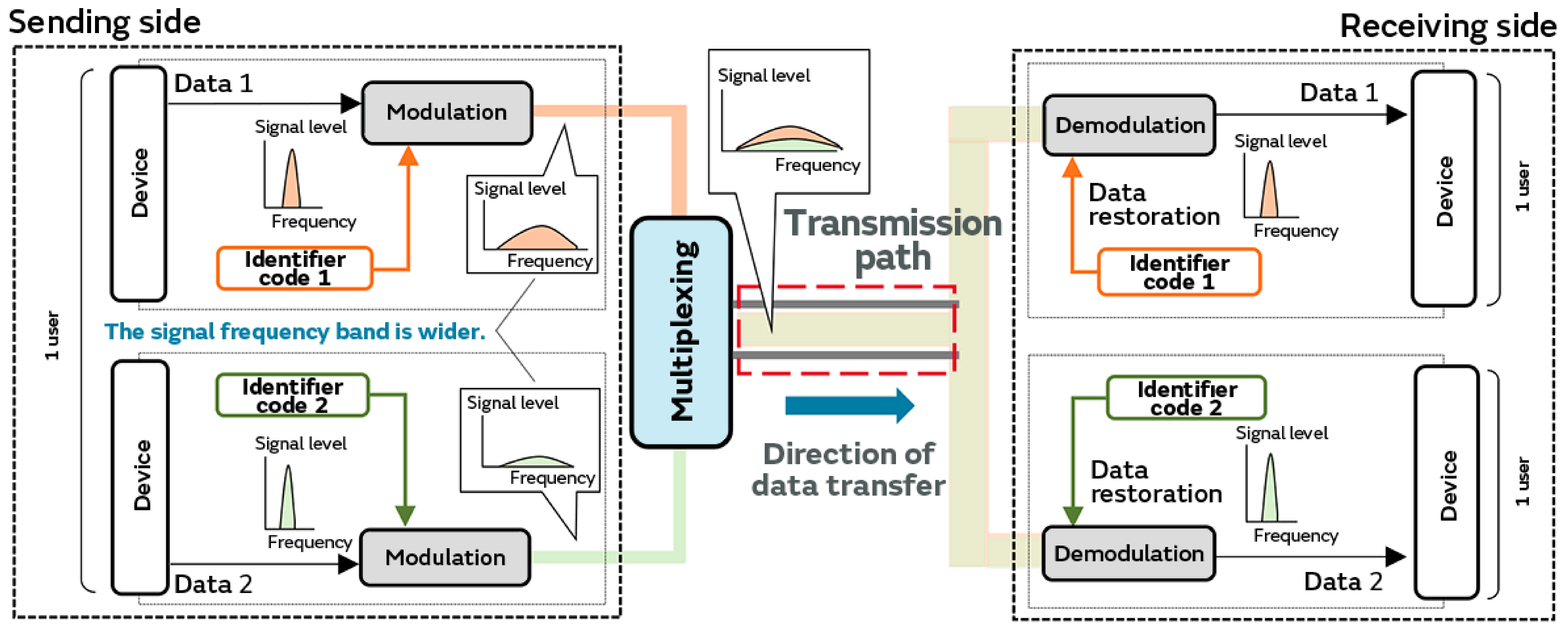


Space Division Multiplexing



(a) Multicore fiber and SDM

Code Division Multiplexing



Modulated signals are generated by mixing each data stream with a unique identifier code, and multiplexing is achieved by adding together these modulated signals.

The modulated signals are mixed with the identifier codes (demodulation), and only the data that matches the specific identifier code is restored.

Advantages and Disadvantages

OF MULTIPLEXING

Advantages

Efficient Use of Bandwidth, Increased Data Transmission, Scalability, Flexibility

Disadvantages

Synchronization Issues, Latency, Signal Degradation, Resource Management

THANK YOU

FOR LISTENING